TECHNISCHE UNIVERSITÄT MÜNCHEN

Quantum Chemistry Laboratory

Prof. N. Rösch

Full Solution of Exercise 17

b. **Optimized distances of $Ni(CO)_4$ in Å**

method/basis	Ni-C	C-O
HF/STO-3G	1.582	1.160
HF/LanL2MB	1.982	1.148
LSDA/LanL2MB	1.856	1.199
Experiment[a]	1.82	1.15

[a]L. Hedberg, T. Iijima, and K. Hedberg, *J. Chem. Phys.* **70** (1979) 3224.

As a late transition metal Ni(0) is an electron-rich atom and therefore electron correlation has to be taken into account. The Hartree-Fock results for the Ni-C bond lengths are very poor because electron correlation is lacking in this method. LSDA provides better results and is recommended especially for cost efficient geometry optimizations of an organometallic compound involving late transition metal atoms.

c. Z-matrices of nickel tricarbonyl and CO with optimized parameters

d. **FBDE in kJ/mol**

LSDA/LanL2MB	222
B3LYP/LanL2DZ at LSDA/LanL2MB geometry	98

Experiment[a]	105 ± 8

[a]A. E. Stevens, C. S. Feigerle, and W. C. Lineberger, *J. Am. Chem. Soc.* **104** (1982) 5026.



[General help]



[Overview]



[Compute]



[Contact the course supervisor]

Copyright © 2003 N. Rösch, Technische Universität München